# WHAT HAVE YOU LEARNED?

Refer to your TABLE OF COMMON IONS and try the following

- How many electrons does a hydrogen atom lose to become an ion? _____
- How many electrons does an oxygen atom gain to become ionized? _____
- Does a zinc atom gain or lose electrons to become ionized? _____
- If an atom commonly loses three electrons to become ionized. Which element could it belong to? _____
- The formula of copper II oxide is CuO. How many electrons are transferred from the copper to the oxygen atoms? _____
- Explain why *two* sodium ions are needed to combine with *one* sulfate ion to form the ionic solid sodium sulfate, $Na_2SO_4$.

_____

_____

- Explain why *two* nitrate ions are needed to combine with *one* lead ion to form the ionic solid lead nitrate, $Pb(NO_3)_2$

_____

_____

- Circle the polyatomic ions $Ba^{2+}$, $OH^-$, $CO_3^{2-}$, $Cu^+$, $Fe^{3+}$, $SO_4^{2-}$, $F^-$, $CH_3COO^-$, $Ag^+$, $PO_4^{3-}$.
- *Without looking* at your TABLE of COMMON IONS, write in the formula of each of the following ions copper I _____, hydroxide _____, lead _____, calcium _____, iodine _____, barium _____, oxide _____, carbonate _____.
- Name these compounds

| | |
|---|---|
| $CuSO_4$ _____ | $Al_2O_3$ _____ |
| $AgCl$ _____ | $Na_2SO_4$ _____ |
| $Ba(OH)_2$ _____ | $PbI_2$ _____ |
| $NH_4NO_3$ _____ | $KCl$ _____ |
| $CaCO_3$ _____ | $NaCl$ _____ |

Now try writing formula (remember to balance the ionic charges)

potassium iodide _____
potassium permanganate _____
copper II sulfate _____
lead nitrate _____
sodium carbonate _____

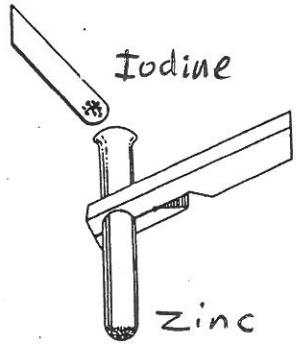
# REACT A METAL WITH A NON-METAL

CARRY OUT THE REACTION OF ZINC WITH IODINE

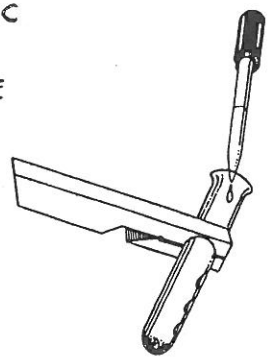
PUT SAFETY GLASSES ON !

1. Collect a test tube that already has some zinc in it.

2. Add an equal quantity of Iodine to the zinc




3. Using a dropper carefully add two or three drops of distilled water to the test-tube. Observe any changes. Feel the bottom of the test-tube.



Complete this - Zinc + Iodine →
(metal) + ( ) →

4. When the reaction has stopped, add about 3 cm of distilled water to the test-tube and shake it gently. Allow the contents of the test-tube to settle.

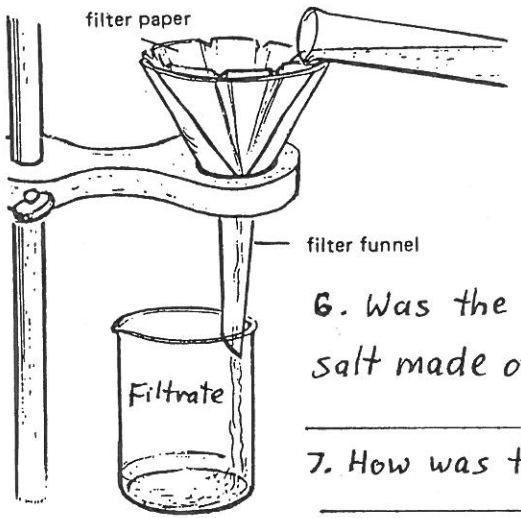
(a) *Describe what you observed during this reaction.*

(b) *How do you know that a reaction has taken place?*

(c) *What is the grey solid left on the bottom of the test-tube?*

(d) *How can you test whether a soluble substance (salt) has been produced?*

5. Filter the solution and keep the filtrate for testing




6. Was the soluble salt made of ions?

7. How was this proven?